

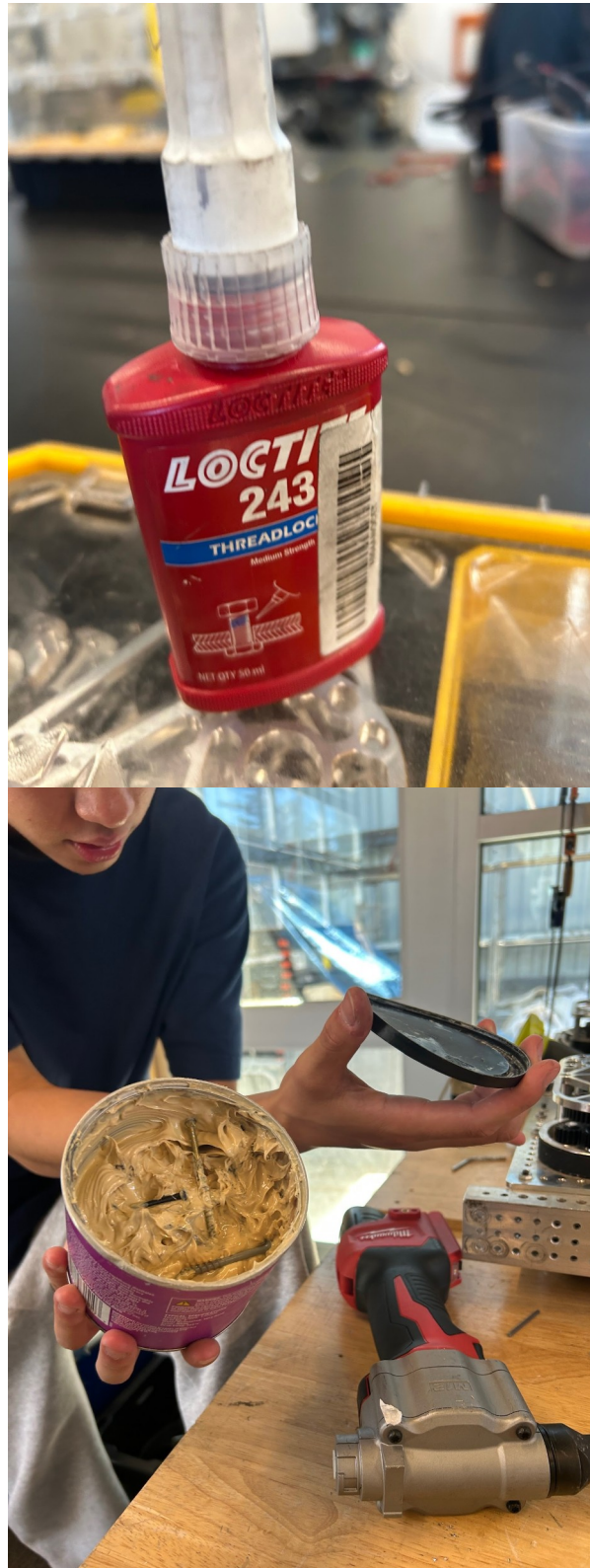
Name: [Arnav Krishnan](#)

Report Date: Week of August 17th

Work Completion Conflicts: None

Completed Tasks

General Assembly Lessons - Monday - 50min

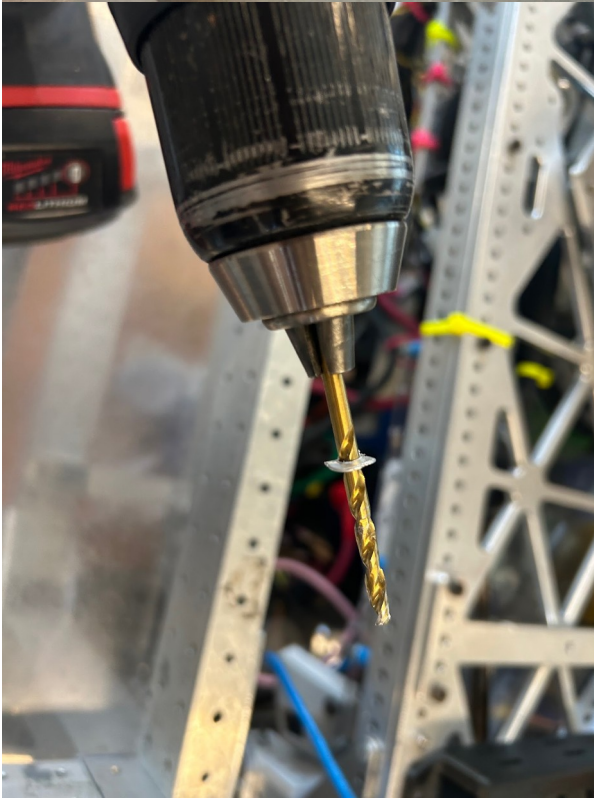


- For today, I was with the Assembly group (Ethan). We focused on the foundational aspects of assembly, such as maintenance techniques and hardware selection, which are very critical when it comes to building a functional and reliable robot.
- We covered using Loctite as a fastener retention method, specifically a thread-locker. Since some screws do not have nuts on the end, and many of these are absolutely critical to the robot, they must be secured with Loctite. We covered the different grades of Loctite, such as blue, the standard one we have and will be using for Robotics; green, which there is significantly less of, and is stronger; and finally red, which should essentially never be used, and should be completely removed from the robotics room.
 - We learned that we cannot rely on the color of the label on the outside of the Loctite bottle. Although the word “THREADLOCK” appears to have a blue background or highlight, this is not indicative of the strength, nor color, of the Loctite. This is an important consideration I will have to be mindful of.
- We also learned about proper lubrication, and noted that standard oils, such as WD-40, should not be used as a primary lubricant. Instead, we use a grease (I believe it is colloquially referred to as “strawberry jam” or “jelly”) for the gears.
 - It will be important to remember to grease gears to increase the lifespan of the robot, and its parts.
- Finally, we went over the batteries available to our team, distinguishing between competition and practice batteries. This is to prevent cell degradation and voltage drop, which

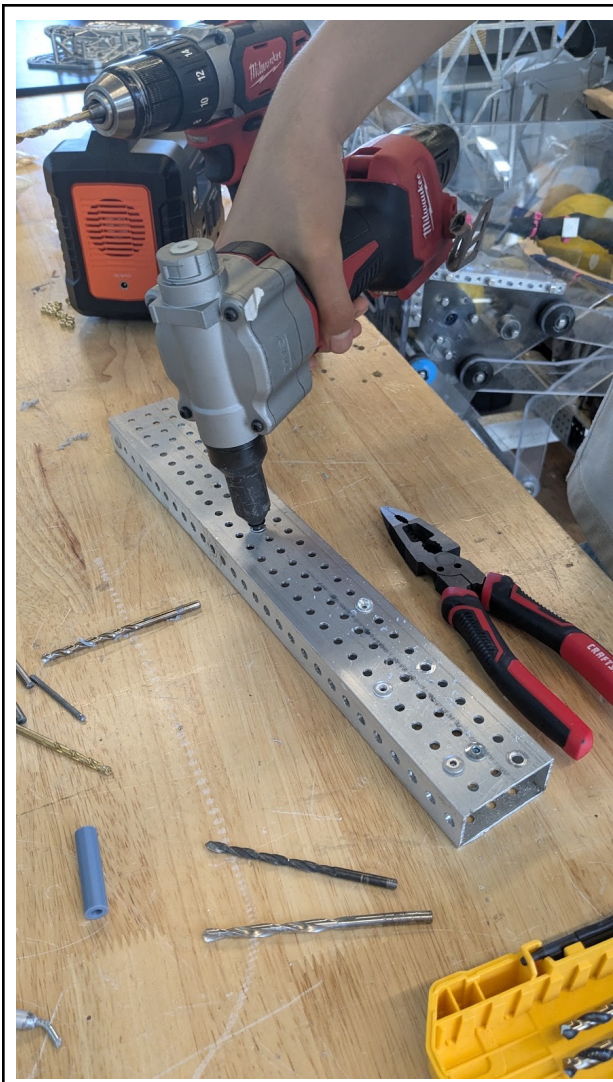


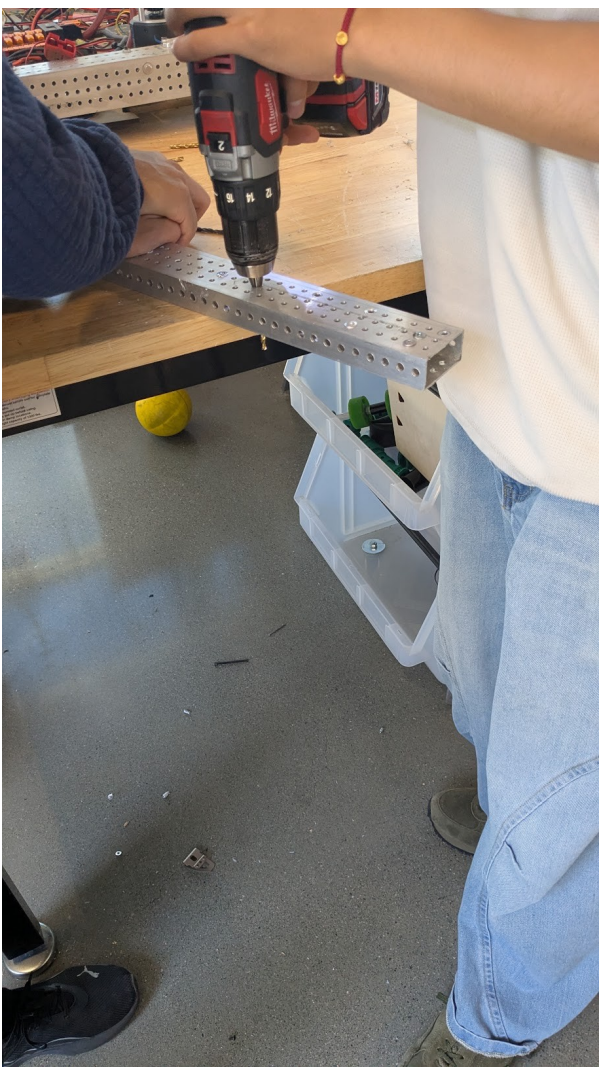
is a significant concern, especially come competition time.

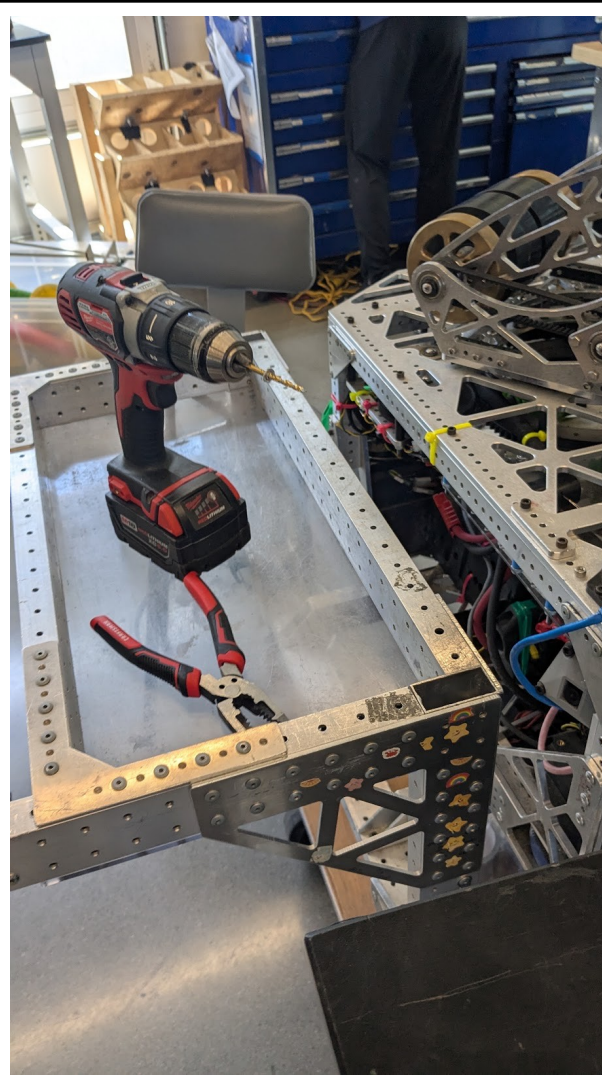
Learning how to rivet and remove rivets - Wednesday - 1hr



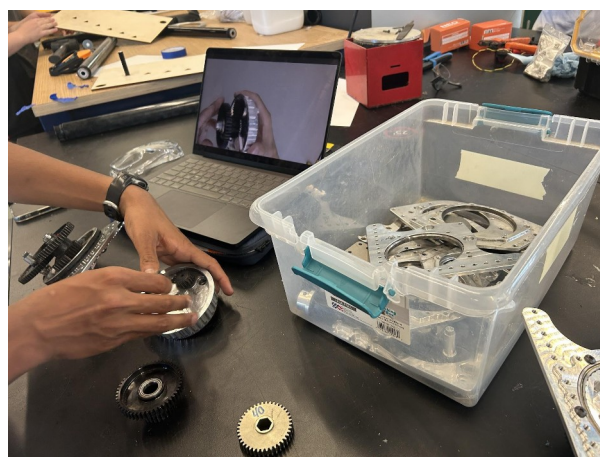
- We focused on the pneumatic riveting process. I learned about the mechanical process behind the pop rivet, and the specific tools, steps, and PPE needed to install (and extract) a rivet successfully.
- We ensured that we were familiar with how to choose the correct drill bit diameter to drill out the center piece of the rivet, without causing the rivet to expand, or otherwise damaging the parent hole.
- One primary challenge we faced is that rivets are often not placed neatly on a piece of metal with nice holes in them; often, they are locked in tight areas that necessitate specialized techniques.
 - However, when working, especially in a confined space, the risk of an off-angle rivet gun placement is drastically increased, and in this case one needs to be much more careful, Ethan showed us how to do this.
 - This showed me that fastening techniques, especially those for rivets, did not just require a basic mechanical understanding of the physical mechanism behind the gun, but rather also a good understanding of how/when to use them
- This was a shorter task, as the rest of the period was spent on the swerve drive (as well as the next day)-- hence the shorter response.



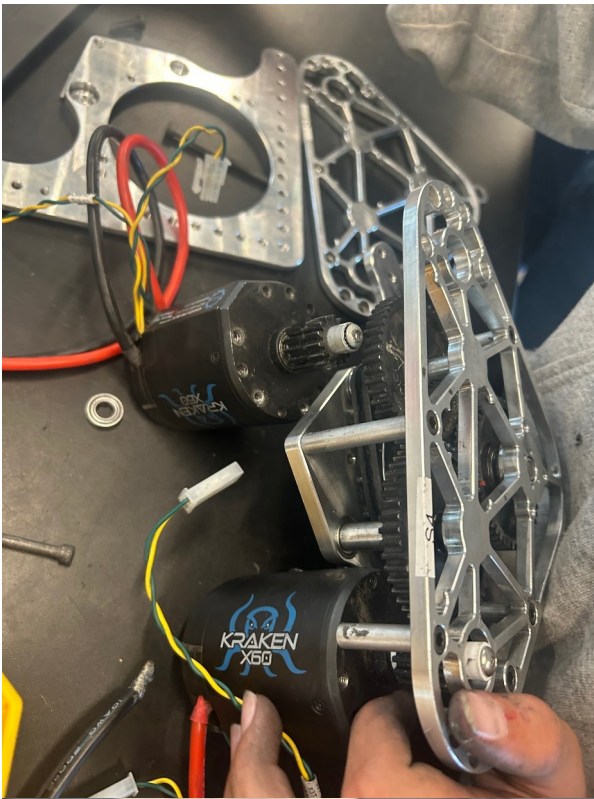




Swerve Drive Assembly - Wednesday (end) + Friday - 2hrs



- We started to learn about why swerve drives are considered the 'gold-standard' in robotics. A huge factor contributing to any robots' success is the swerve drive.
- We learned the process of putting a swerve drive together, as we followed along to a youtube video
- One thing I noticed is the tread that is secured around the wheels does have 2 screws as fasteners, which results in the wheel not being perfectly uniform. I expect that the effect of the screw is very small, because the screw head is so short, but I am curious if it causes the robot to lose traction in certain wheels at a certain time, periodically.
- We learned about how to apply the "strawberry jam" grease, which, if I recall correctly, is lithium grease, in order to reduce the amount of decay experienced by the teeth in the gearbox.
- We saw why it was so important that



metal-to-metal screws without nuts on the other end receive Loctite.

- We should ensure we are only using Blue loctite
- 1-3 threads on the screw
- We also use different motors, specifically “Krakens”, which are much more powerful than the ones we used in Robotics I.
- Finally, we ensured that we cleaned up all the materials and organized them, as keeping the shop organized and clean is important not just for ease of locating items but also safety.

